

The following table provides the complete enumeration of the 106 descriptors comprising the AutoAIM Custom featurizer. The featurizer encodes 13 elemental properties through eight statistical moments (mean, std, min, max, range, median, p25, p75), supplemented by two composition-derived features (n_elements, fractional_entropy), totaling 106 descriptors. All elemental data are drawn from standard tabulated sources (Mendelev database, CRC Handbook of Chemistry and Physics) embedded within the AutoAIM software.

(Part 1/5). Elemental Properties (40 descriptors = 5 properties x 8 statistics).

Category	Descriptor Name	Definition	Source
Elemental	comp_atomic_number_mean	Weighted mean: $\bar{x} = \sum f_i \cdot \text{atomic_number}_i$. Ordinal identifier in the periodic table	Tabulated elemental data (AutoAIM)
Elemental	comp_atomic_number_std	Weighted std: $\sigma = \sqrt{[\sum (x_i - \bar{x})^2 \cdot f_i]}$. Ordinal identifier in the periodic table	Tabulated elemental data (AutoAIM)
Elemental	comp_atomic_number_min	Minimum atomic number among constituent elements. Ordinal identifier in the periodic table	Tabulated elemental data (AutoAIM)
Elemental	comp_atomic_number_max	Maximum atomic number among constituent elements. Ordinal identifier in the periodic table	Tabulated elemental data (AutoAIM)
Elemental	comp_atomic_number_range	Range: $\max(\text{atomic_number}) - \min(\text{atomic_number})$. Ordinal identifier in the periodic table	Tabulated elemental data (AutoAIM)
Elemental	comp_atomic_number_median	Weighted median of atomic number. Ordinal identifier in the periodic table	Tabulated elemental data (AutoAIM)
Elemental	comp_atomic_number_p25	25th percentile (Q1) of atomic number. Ordinal identifier in the periodic table	Tabulated elemental data (AutoAIM)
Elemental	comp_atomic_number_p75	75th percentile (Q3) of atomic number. Ordinal identifier in the periodic table	Tabulated elemental data (AutoAIM)
Elemental	comp_atomic_mass_mean	Weighted mean: $\bar{x} = \sum f_i \cdot \text{atomic_mass}_i$. Atomic weight (g/mol)	Tabulated elemental data (AutoAIM)
Elemental	comp_atomic_mass_std	Weighted std: $\sigma = \sqrt{[\sum (x_i - \bar{x})^2 \cdot f_i]}$. Atomic weight (g/mol)	Tabulated elemental data (AutoAIM)
Elemental	comp_atomic_mass_min	Minimum atomic mass among constituent elements. Atomic weight (g/mol)	Tabulated elemental data (AutoAIM)

Category	Descriptor Name	Definition	Source
Elemental	comp_atomic_mass_max	Maximum atomic mass among constituent elements. Atomic weight (g/mol)	Tabulated elemental data (AutoAIM)
Elemental	comp_atomic_mass_range	Range: max(atomic_mass) - min(atomic_mass). Atomic weight (g/mol)	Tabulated elemental data (AutoAIM)
Elemental	comp_atomic_mass_median	Weighted median of atomic mass. Atomic weight (g/mol)	Tabulated elemental data (AutoAIM)
Elemental	comp_atomic_mass_p25	25th percentile (Q1) of atomic mass. Atomic weight (g/mol)	Tabulated elemental data (AutoAIM)
Elemental	comp_atomic_mass_p75	75th percentile (Q3) of atomic mass. Atomic weight (g/mol)	Tabulated elemental data (AutoAIM)
Elemental	comp_group_mean	Weighted mean: $\bar{x} = \sum f_i \cdot \text{group}_i$. Periodic table column	Tabulated elemental data (AutoAIM)
Elemental	comp_group_std	Weighted std: $\sigma = \sqrt{[\sum (x_i - \bar{x})^2 \cdot f_i]}$. Periodic table column	Tabulated elemental data (AutoAIM)
Elemental	comp_group_min	Minimum group among constituent elements. Periodic table column	Tabulated elemental data (AutoAIM)
Elemental	comp_group_max	Maximum group among constituent elements. Periodic table column	Tabulated elemental data (AutoAIM)
Elemental	comp_group_range	Range: max(group) - min(group). Periodic table column	Tabulated elemental data (AutoAIM)
Elemental	comp_group_median	Weighted median of group. Periodic table column	Tabulated elemental data (AutoAIM)
Elemental	comp_group_p25	25th percentile (Q1) of group. Periodic table column	Tabulated elemental data (AutoAIM)
Elemental	comp_group_p75	75th percentile (Q3) of group. Periodic table column	Tabulated elemental data (AutoAIM)
Elemental	comp_period_mean	Weighted mean: $\bar{x} = \sum f_i \cdot \text{period}_i$. Periodic table row	Tabulated elemental data (AutoAIM)
Elemental	comp_period_std	Weighted std: $\sigma = \sqrt{[\sum (x_i - \bar{x})^2 \cdot f_i]}$. Periodic table row	Tabulated elemental data (AutoAIM)
Elemental	comp_period_min	Minimum period among constituent elements. Periodic table row	Tabulated elemental data (AutoAIM)

Category	Descriptor Name	Definition	Source
Elemental	comp_period_max	Maximum period among constituent elements. Periodic table row	Tabulated elemental data (AutoAIM)
Elemental	comp_period_range	Range: max(period) - min(period). Periodic table row	Tabulated elemental data (AutoAIM)
Elemental	comp_period_median	Weighted median of period. Periodic table row	Tabulated elemental data (AutoAIM)
Elemental	comp_period_p25	25th percentile (Q1) of period. Periodic table row	Tabulated elemental data (AutoAIM)
Elemental	comp_period_p75	75th percentile (Q3) of period. Periodic table row	Tabulated elemental data (AutoAIM)
Elemental	comp_atomic_radius_mean	Weighted mean: $\bar{x} = \sum f_i \cdot \text{atomic_radius}_i$. Van der Waals radius (pm)	Tabulated elemental data (AutoAIM)
Elemental	comp_atomic_radius_std	Weighted std: $\sigma = \sqrt{[\sum (x_i - \bar{x})^2 \cdot f_i]}$. Van der Waals radius (pm)	Tabulated elemental data (AutoAIM)
Elemental	comp_atomic_radius_min	Minimum atomic radius among constituent elements. Van der Waals radius (pm)	Tabulated elemental data (AutoAIM)
Elemental	comp_atomic_radius_max	Maximum atomic radius among constituent elements. Van der Waals radius (pm)	Tabulated elemental data (AutoAIM)
Elemental	comp_atomic_radius_range	Range: max(atomic_radius) - min(atomic_radius). Van der Waals radius (pm)	Tabulated elemental data (AutoAIM)
Elemental	comp_atomic_radius_median	Weighted median of atomic radius. Van der Waals radius (pm)	Tabulated elemental data (AutoAIM)
Elemental	comp_atomic_radius_p25	25th percentile (Q1) of atomic radius. Van der Waals radius (pm)	Tabulated elemental data (AutoAIM)
Elemental	comp_atomic_radius_p75	75th percentile (Q3) of atomic radius. Van der Waals radius (pm)	Tabulated elemental data (AutoAIM)

(Part 2/5). Statistical Moments (24 descriptors = 3 properties x 8 statistics).

Category	Descriptor Name	Definition	Source
Statistical	comp_covalent_radius_mean	Weighted mean: $\bar{x} = \sum f_i \cdot \text{covalent_radius}_i$. Covalent bond radius (pm)	Tabulated elemental data (AutoAIM)

Category	Descriptor Name	Definition	Source
Statistical	comp_covalent_radius_std	Weighted std: $\sigma = \sqrt{[\sum(x_i - \bar{x})^2 \cdot f_i]}$. Covalent bond radius (pm)	Tabulated elemental data (AutoAIM)
Statistical	comp_covalent_radius_min	Minimum covalent radius among constituent elements. Covalent bond radius (pm)	Tabulated elemental data (AutoAIM)
Statistical	comp_covalent_radius_max	Maximum covalent radius among constituent elements. Covalent bond radius (pm)	Tabulated elemental data (AutoAIM)
Statistical	comp_covalent_radius_range	Range: max(covalent_radius) - min(covalent_radius). Covalent bond radius (pm)	Tabulated elemental data (AutoAIM)
Statistical	comp_covalent_radius_median	Weighted median of covalent radius. Covalent bond radius (pm)	Tabulated elemental data (AutoAIM)
Statistical	comp_covalent_radius_p25	25th percentile (Q1) of covalent radius. Covalent bond radius (pm)	Tabulated elemental data (AutoAIM)
Statistical	comp_covalent_radius_p75	75th percentile (Q3) of covalent radius. Covalent bond radius (pm)	Tabulated elemental data (AutoAIM)
Statistical	comp_melting_point_mean	Weighted mean: $\bar{x} = \sum f_i \cdot \text{melting_point}_i$. Melting temperature at 1 atm (K)	Tabulated elemental data (AutoAIM)
Statistical	comp_melting_point_std	Weighted std: $\sigma = \sqrt{[\sum(x_i - \bar{x})^2 \cdot f_i]}$. Melting temperature at 1 atm (K)	Tabulated elemental data (AutoAIM)
Statistical	comp_melting_point_min	Minimum melting point among constituent elements. Melting temperature at 1 atm (K)	Tabulated elemental data (AutoAIM)
Statistical	comp_melting_point_max	Maximum melting point among constituent elements. Melting temperature at 1 atm (K)	Tabulated elemental data (AutoAIM)
Statistical	comp_melting_point_range	Range: max(melting_point) - min(melting_point). Melting temperature at 1 atm (K)	Tabulated elemental data (AutoAIM)
Statistical	comp_melting_point_median	Weighted median of melting point. Melting temperature at 1 atm (K)	Tabulated elemental data (AutoAIM)
Statistical	comp_melting_point_p25	25th percentile (Q1) of melting point. Melting temperature at 1 atm (K)	Tabulated elemental data (AutoAIM)
Statistical	comp_melting_point_p75	75th percentile (Q3) of melting point. Melting temperature at 1 atm (K)	Tabulated elemental data (AutoAIM)

Category	Descriptor Name	Definition	Source
Statistical	comp_boiling_point_mean	Weighted mean: $\bar{x} = \sum f_i \cdot \text{boiling_point}_i$. Boiling temperature at 1 atm (K)	Tabulated elemental data (AutoAIM)
Statistical	comp_boiling_point_std	Weighted std: $\sigma = \sqrt{[\sum (x_i - \bar{x})^2 \cdot f_i]}$. Boiling temperature at 1 atm (K)	Tabulated elemental data (AutoAIM)
Statistical	comp_boiling_point_min	Minimum boiling point among constituent elements. Boiling temperature at 1 atm (K)	Tabulated elemental data (AutoAIM)
Statistical	comp_boiling_point_max	Maximum boiling point among constituent elements. Boiling temperature at 1 atm (K)	Tabulated elemental data (AutoAIM)
Statistical	comp_boiling_point_range	Range: $\max(\text{boiling_point}) - \min(\text{boiling_point})$. Boiling temperature at 1 atm (K)	Tabulated elemental data (AutoAIM)
Statistical	comp_boiling_point_median	Weighted median of boiling point. Boiling temperature at 1 atm (K)	Tabulated elemental data (AutoAIM)
Statistical	comp_boiling_point_p25	25th percentile (Q1) of boiling point. Boiling temperature at 1 atm (K)	Tabulated elemental data (AutoAIM)
Statistical	comp_boiling_point_p75	75th percentile (Q3) of boiling point. Boiling temperature at 1 atm (K)	Tabulated elemental data (AutoAIM)

(Part 3/5). Electronic Structure (24 descriptors = 3 properties x 8 statistics).

Category	Descriptor Name	Definition	Source
Electronic	comp_electronegativity_mean	Weighted mean: $\bar{x} = \sum f_i \cdot \text{electronegativity}_i$. Pauling electronegativity scale	Tabulated elemental data (AutoAIM)
Electronic	comp_electronegativity_std	Weighted std: $\sigma = \sqrt{[\sum (x_i - \bar{x})^2 \cdot f_i]}$. Pauling electronegativity scale	Tabulated elemental data (AutoAIM)
Electronic	comp_electronegativity_min	Minimum electronegativity among constituent elements. Pauling electronegativity scale	Tabulated elemental data (AutoAIM)
Electronic	comp_electronegativity_max	Maximum electronegativity among constituent elements. Pauling electronegativity scale	Tabulated elemental data (AutoAIM)

Category	Descriptor Name	Definition	Source
Electronic	comp_electronegativity_range	Range: max(electronegativity) - min(electronegativity). Pauling electronegativity scale	Tabulated elemental data (AutoAIM)
Electronic	comp_electronegativity_median	Weighted median of electronegativity. Pauling electronegativity scale	Tabulated elemental data (AutoAIM)
Electronic	comp_electronegativity_p25	25th percentile (Q1) of electronegativity. Pauling electronegativity scale	Tabulated elemental data (AutoAIM)
Electronic	comp_electronegativity_p75	75th percentile (Q3) of electronegativity. Pauling electronegativity scale	Tabulated elemental data (AutoAIM)
Electronic	comp_ionization_energy_mean	Weighted mean: $\bar{x} = \sum f_i \cdot \text{ionization_energy}_i$. First ionization energy (kJ/mol)	Tabulated elemental data (AutoAIM)
Electronic	comp_ionization_energy_std	Weighted std: $\sigma = \sqrt{[\sum (x_i - \bar{x})^2 \cdot f_i]}$. First ionization energy (kJ/mol)	Tabulated elemental data (AutoAIM)
Electronic	comp_ionization_energy_min	Minimum ionization energy among constituent elements. First ionization energy (kJ/mol)	Tabulated elemental data (AutoAIM)
Electronic	comp_ionization_energy_max	Maximum ionization energy among constituent elements. First ionization energy (kJ/mol)	Tabulated elemental data (AutoAIM)
Electronic	comp_ionization_energy_range	Range: max(ionization_energy) - min(ionization_energy). First ionization energy (kJ/mol)	Tabulated elemental data (AutoAIM)
Electronic	comp_ionization_energy_median	Weighted median of ionization energy. First ionization energy (kJ/mol)	Tabulated elemental data (AutoAIM)
Electronic	comp_ionization_energy_p25	25th percentile (Q1) of ionization energy. First ionization energy (kJ/mol)	Tabulated elemental data (AutoAIM)

Category	Descriptor Name	Definition	Source
Electronic	comp_ionization_energy_p75	75th percentile (Q3) of ionization energy. First ionization energy (kJ/mol)	Tabulated elemental data (AutoAIM)
Electronic	comp_electron_affinity_mean	Weighted mean: $\bar{x} = \sum f_i \cdot \text{electron_affinity}_i$. Electron affinity (kJ/mol)	Tabulated elemental data (AutoAIM)
Electronic	comp_electron_affinity_std	Weighted std: $\sigma = \sqrt{[\sum (x_i - \bar{x})^2 \cdot f_i]}$. Electron affinity (kJ/mol)	Tabulated elemental data (AutoAIM)
Electronic	comp_electron_affinity_min	Minimum electron affinity among constituent elements. Electron affinity (kJ/mol)	Tabulated elemental data (AutoAIM)
Electronic	comp_electron_affinity_max	Maximum electron affinity among constituent elements. Electron affinity (kJ/mol)	Tabulated elemental data (AutoAIM)
Electronic	comp_electron_affinity_range	Range: $\max(\text{electron_affinity}) - \min(\text{electron_affinity})$. Electron affinity (kJ/mol)	Tabulated elemental data (AutoAIM)
Electronic	comp_electron_affinity_median	Weighted median of electron affinity. Electron affinity (kJ/mol)	Tabulated elemental data (AutoAIM)
Electronic	comp_electron_affinity_p25	25th percentile (Q1) of electron affinity. Electron affinity (kJ/mol)	Tabulated elemental data (AutoAIM)
Electronic	comp_electron_affinity_p75	75th percentile (Q3) of electron affinity. Electron affinity (kJ/mol)	Tabulated elemental data (AutoAIM)

(Part 4/5). Thermodynamic Properties (16 descriptors = 2 properties x 8 statistics).

Category	Descriptor Name	Definition	Source
Thermodynamic	comp_density_mean	Weighted mean: $\bar{x} = \sum f_i \cdot \text{density}_i$. Mass density at STP (g/cm ³)	Tabulated elemental data (AutoAIM)

Category	Descriptor Name	Definition	Source
Thermodynamic	comp_density_std	Weighted std: $\sigma = \sqrt{[\sum(x_i - \bar{x})^2 \cdot f_i]}$. Mass density at STP (g/cm ³)	Tabulated elemental data (AutoAIM)
Thermodynamic	comp_density_min	Minimum density among constituent elements. Mass density at STP (g/cm ³)	Tabulated elemental data (AutoAIM)
Thermodynamic	comp_density_max	Maximum density among constituent elements. Mass density at STP (g/cm ³)	Tabulated elemental data (AutoAIM)
Thermodynamic	comp_density_range	Range: max(density) - min(density). Mass density at STP (g/cm ³)	Tabulated elemental data (AutoAIM)
Thermodynamic	comp_density_median	Weighted median of density. Mass density at STP (g/cm ³)	Tabulated elemental data (AutoAIM)
Thermodynamic	comp_density_p25	25th percentile (Q1) of density. Mass density at STP (g/cm ³)	Tabulated elemental data (AutoAIM)
Thermodynamic	comp_density_p75	75th percentile (Q3) of density. Mass density at STP (g/cm ³)	Tabulated elemental data (AutoAIM)
Thermodynamic	comp_thermal_conductivity_mean	Weighted mean: $\bar{x} = \sum f_i \cdot \text{thermal_conductivity}_i$. Thermal conductivity at 300 K (W/m·K)	Tabulated elemental data (AutoAIM)
Thermodynamic	comp_thermal_conductivity_std	Weighted std: $\sigma = \sqrt{[\sum(x_i - \bar{x})^2 \cdot f_i]}$. Thermal conductivity at 300 K (W/m·K)	Tabulated elemental data (AutoAIM)
Thermodynamic	comp_thermal_conductivity_min	Minimum thermal conductivity among constituent elements. Thermal conductivity at 300 K (W/m·K)	Tabulated elemental data (AutoAIM)
Thermodynamic	comp_thermal_conductivity_max	Maximum thermal conductivity among constituent elements. Thermal conductivity at 300 K (W/m·K)	Tabulated elemental data (AutoAIM)

Category	Descriptor Name	Definition	Source
Thermodynamic	comp_thermal_conductivity_range	Range: max(thermal_conductivity) - min(thermal_conductivity). Thermal conductivity at 300 K (W/m·K)	Tabulated elemental data (AutoAIM)
Thermodynamic	comp_thermal_conductivity_median	Weighted median of thermal conductivity. Thermal conductivity at 300 K (W/m·K)	Tabulated elemental data (AutoAIM)
Thermodynamic	comp_thermal_conductivity_p25	25th percentile (Q1) of thermal conductivity. Thermal conductivity at 300 K (W/m·K)	Tabulated elemental data (AutoAIM)
Thermodynamic	comp_thermal_conductivity_p75	75th percentile (Q3) of thermal conductivity. Thermal conductivity at 300 K (W/m·K)	Tabulated elemental data (AutoAIM)

(Part 5/5). Composition Ratios (2 descriptors).

Category	Descriptor Name	Definition	Source
Composition	comp_n_elements	Number of unique constituent elements in the chemical formula	Derived from composition
Composition	comp_fractional_entropy	Shannon compositional entropy: $H = -\sum f_i \cdot \ln(f_i)$ where f_i is the atomic fraction of element i	Derived from composition